



SBI for Simple Model of Neuron Activation

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Motivations

- Tiny brain cells communicate via electrical signals.
- Parameters are key but hard to measure directly.
- Estimate these parameters from voltage and current recordings using simulation - based Bayesian inference

The LIF Model and Its Parameters

$$\tau_m \frac{\partial V}{\partial t} = -(V - V_{rest}) + RI$$

- V : Membrane potential, V_{rest} : Membrane potential in resting state
- I : Input current
- R : Membrane Resistance
- τ_m : Time constant

Voltage V increases, spikes when threshold V_{th} is hit, then resets to V_{rest}

Example

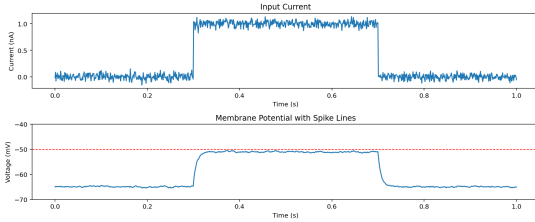


Figure: Voltage without spikes

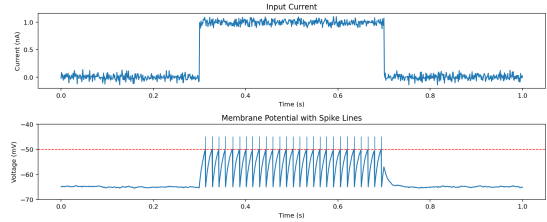


Figure: Voltage with spikes

Bayesian Framework

$$\tau_m \frac{\partial V}{\partial t} = -(V - V_{\text{rest}}) + RI$$

with

$$P(\theta \mid y) = \frac{P(y \mid \theta) \cdot P(\theta)}{P(y)}$$

- Parameters: $\theta = (\tau_m, R)$
- Observations: $y = V(t)$

Simulator

- Specific capacitance: $c_m \sim \text{LogNormal}(\log(10), 0.05^2)$
- Specific resistance: $r_m \sim \text{LogNormal}(\log(1), 0.5^2)$
- Surface area: $A = 0.05$

Parameters

- $\tau_m = c_m \cdot r_m$
- $R = r_m / A$

Simulator

- Keep V_{th} and V_{rest} fixed
- I as rectangular pulse with high state 1,
- Noisy in I : $\mathcal{N}(0, 0.05^2)$

Observations

- $V(t) = \text{LIF}(\tau_m, R, \dots)$

Approximator

1.st Version

- Adapter: I , V and spikes as conditional Variables
- Summary Network: LSTM
- few epochs: about 15 to 30

2.nd Version

- Adapter: only V as conditional Variables, standardized τ_m and R
- Summary Network: sequence of dense layers
- few epochs: about 500 to 1000

Diagnostics - 1.st Version

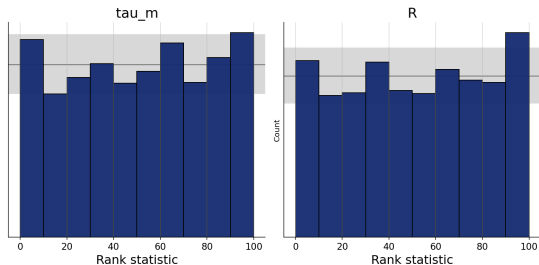


Figure: SBC Histogramms - 1.st Approximator

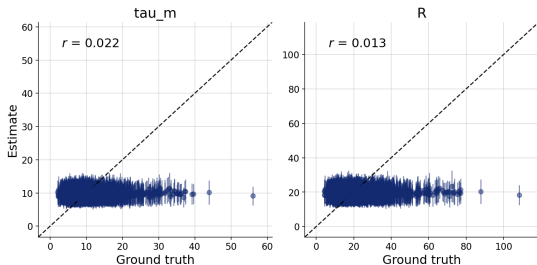


Figure: Parameters Recovery - 1.st Approximator

Diagnostics - 1.st Version

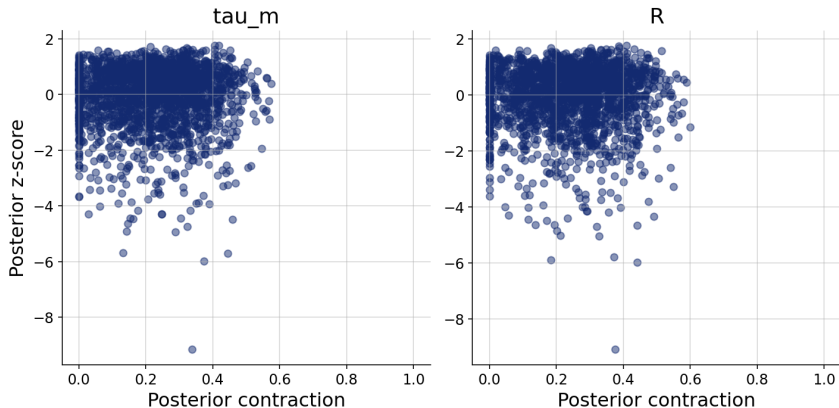


Figure: Z-score vs. contraction plot - 1.st Approximator

Diagnostics - 2.nd Version

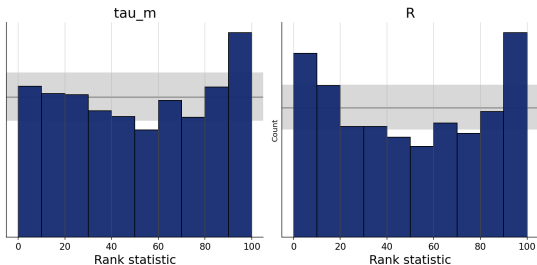


Figure: SBC Histogramms - 2.nd Approximator

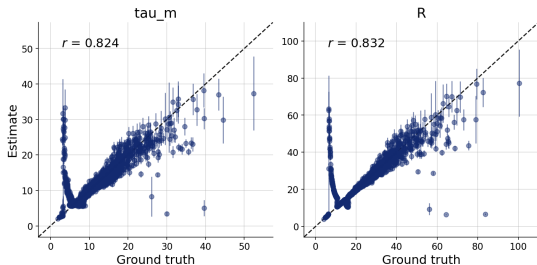


Figure: Parameters Recovery - 2.nd Approximator

Diagnostics - 2.nd Version

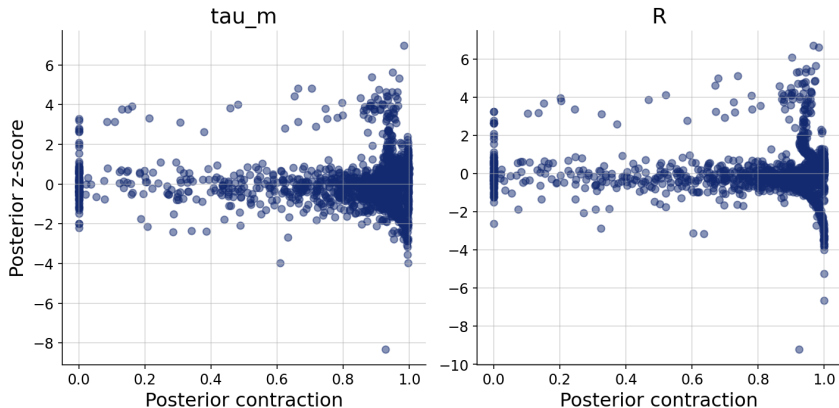


Figure: Z-score vs. contraction plot - 2.nd Approximator

Diskussionsfragen

- Consider model complexity to balance training budget.
- Simple data processing can have a big impact. (ex. Standardization)
- Debugging is complicated and requires deeper understanding.